



UNIMORE
UNIVERSITÀ DEGLI STUDI DI
MODENA E REGGIO EMILIA

Dipartimento di Ingegneria
"Enzo Ferrari"

Processi di asportazione di truciolo

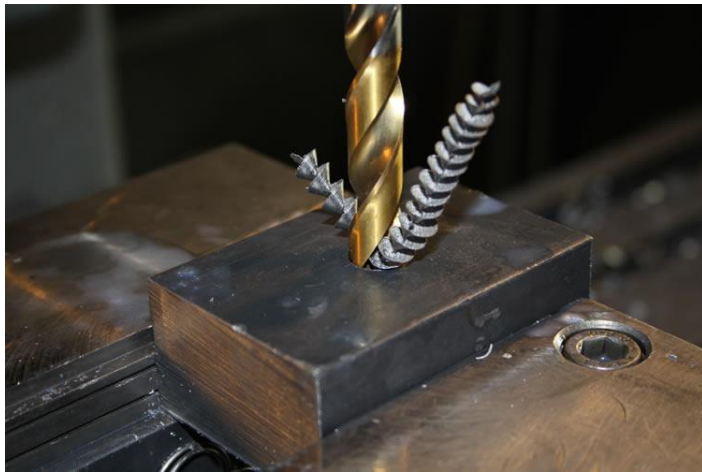
Silvio Defanti
silvio.defanti@unimore.it

Introduzione

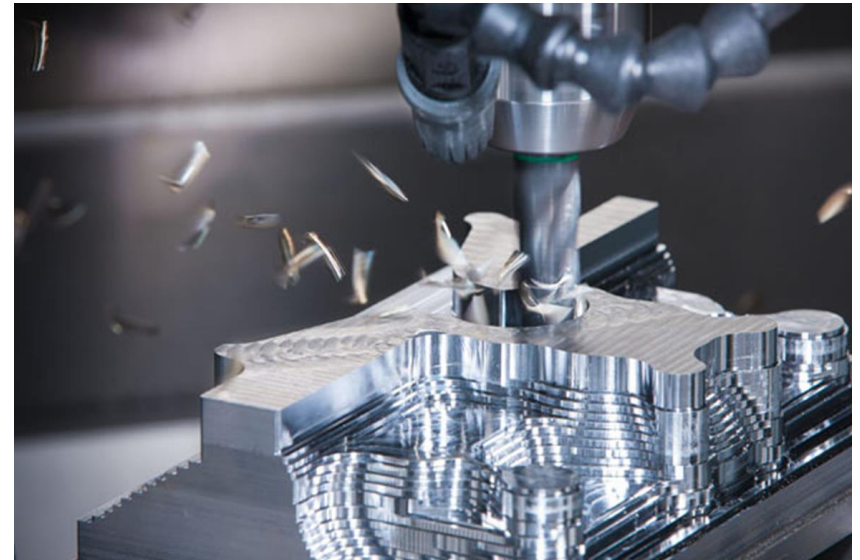
Tecniche di **lavorazione secondaria** che asportano materiale da un semilavorato per ottenere la geometria precisa del prodotto finale. Si applicano soprattutto a materiali metallici e per piccoli lotti.



Tornitura



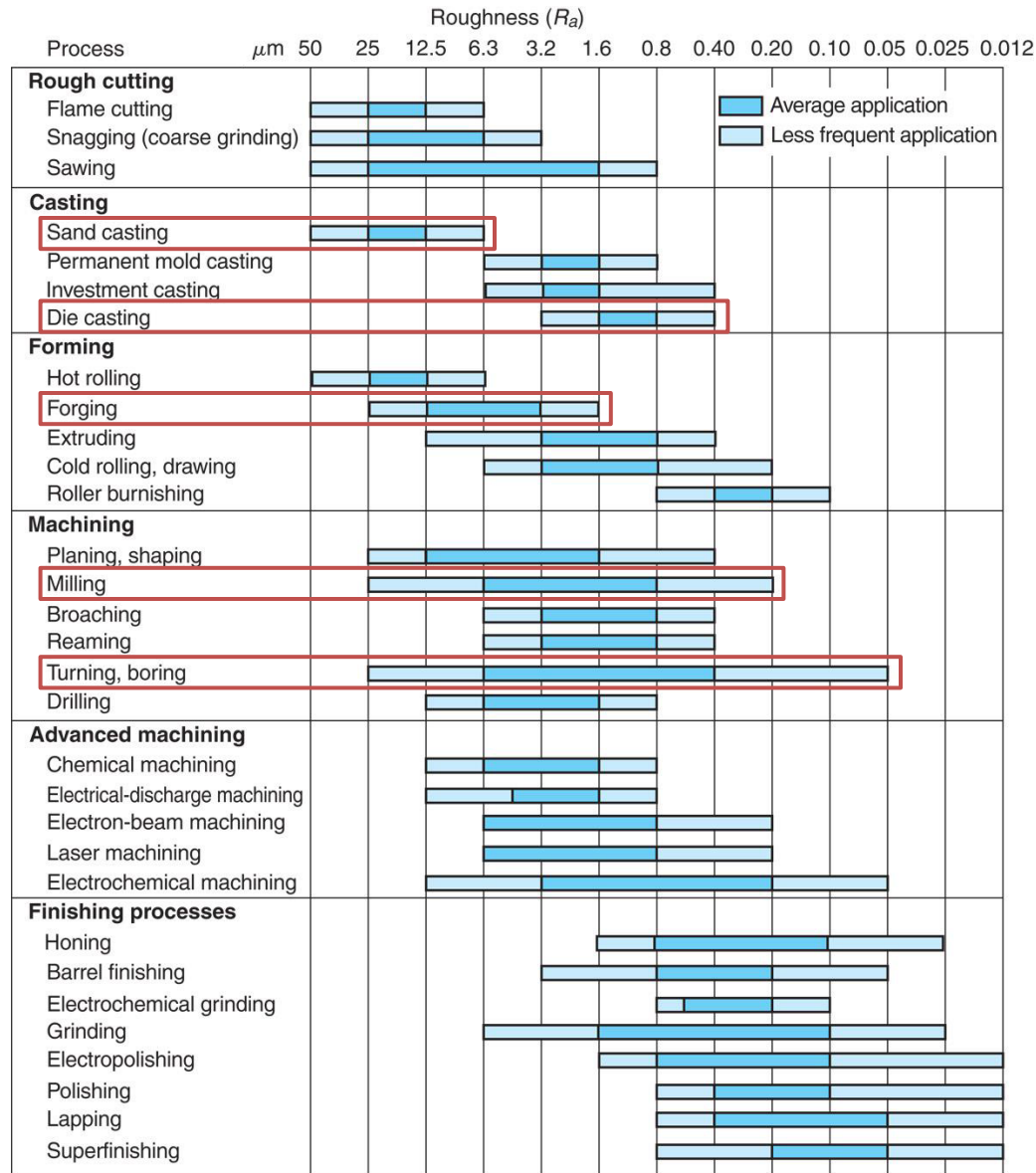
Foratura



Fresatura

[Video: lavorazione di un prodotto ottenuto per fusione](#)

Introduzione



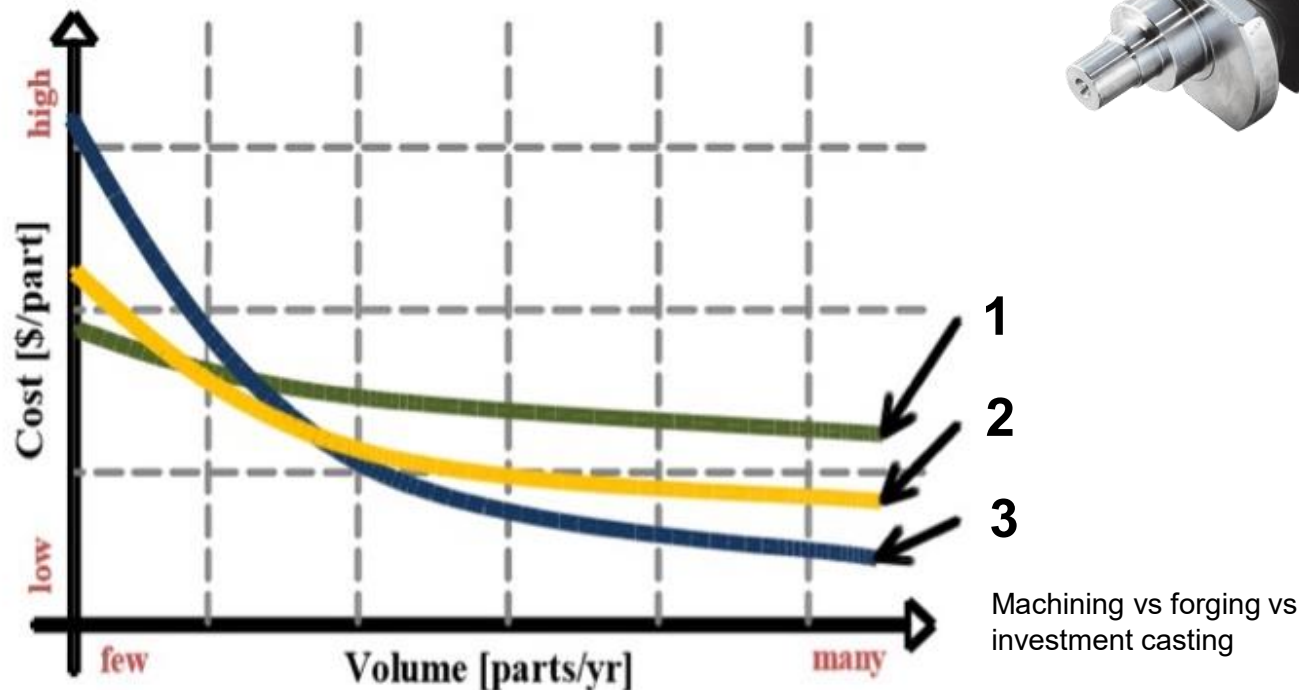
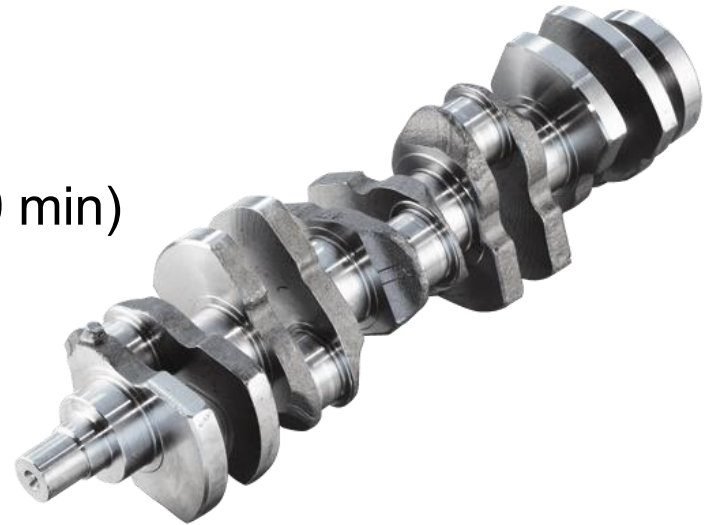
Introduzione

Realizzazione di un albero motore

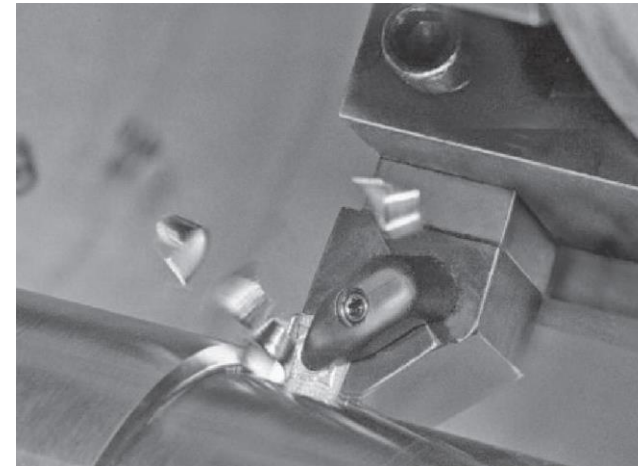
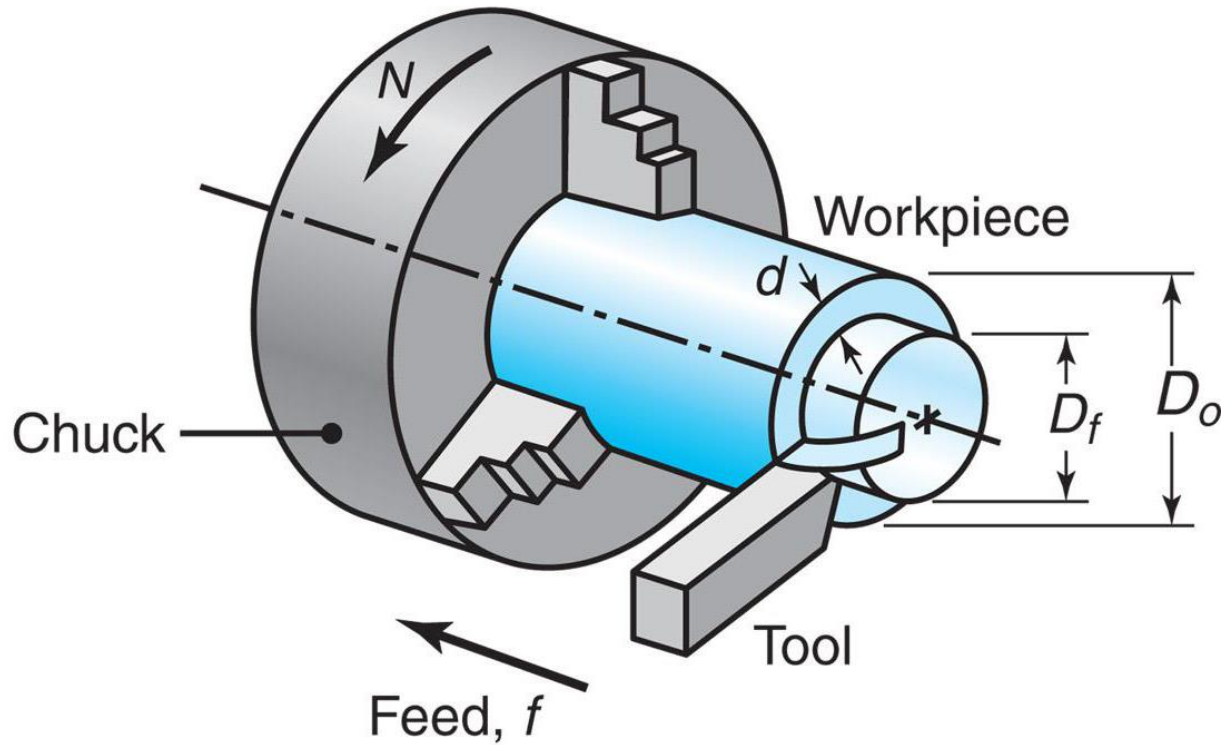
Opzione A) Tornitura (200 ore)

Opzione B) Forgiatura (2 ore) + tornitura (20 min)

Quale conviene?

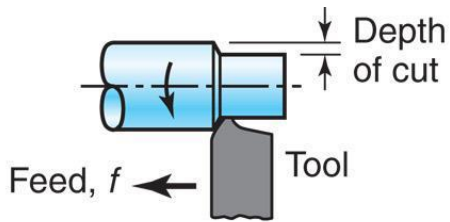


Tornitura

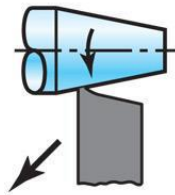


[Video: tornitura](#)

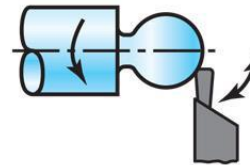
Tornitura



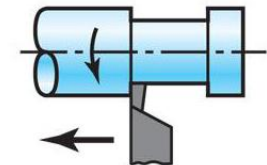
(a) Straight turning



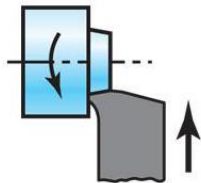
(b) Taper turning



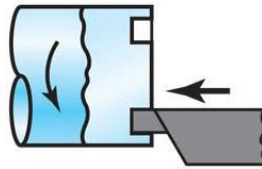
(c) Profiling



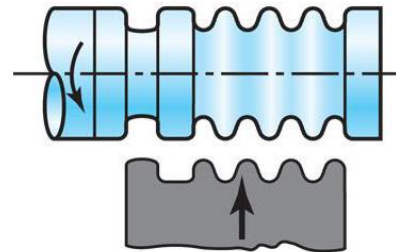
(d) Turning and external grooving



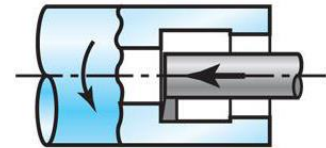
(e) Facing



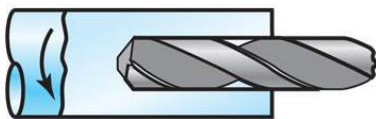
(f) Face grooving



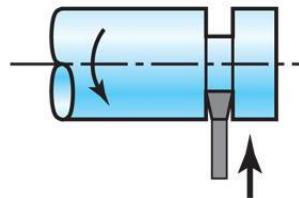
(g) Cutting with a form tool



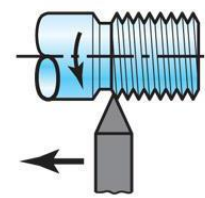
(h) Boring and internal grooving



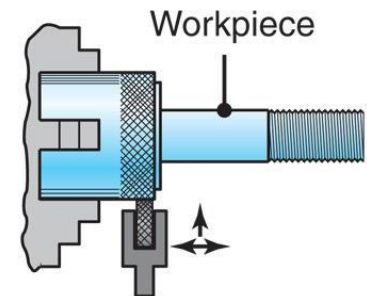
(i) Drilling



(j) Cutting off



(k) Threading



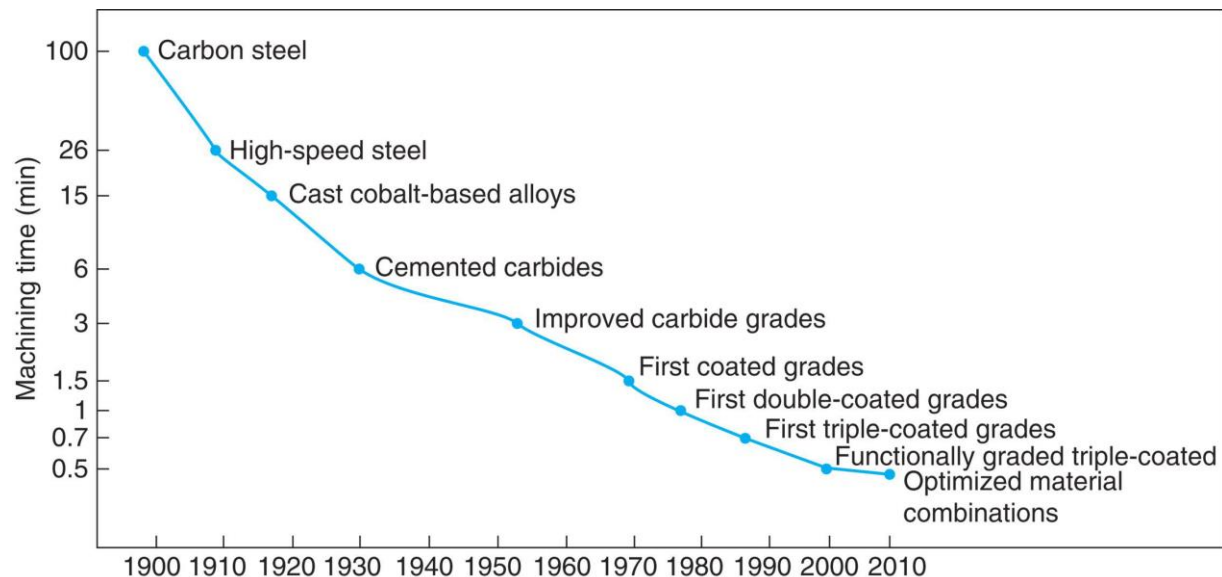
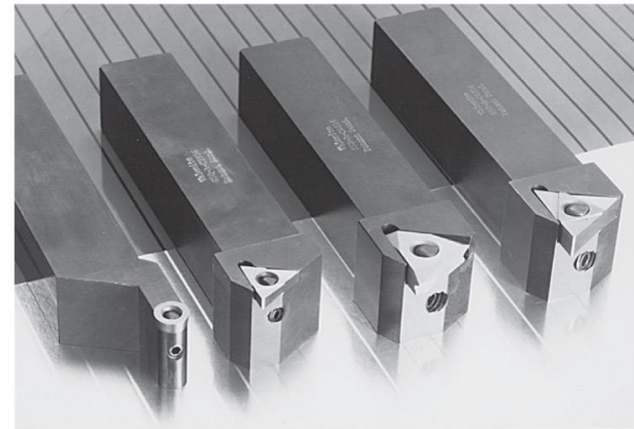
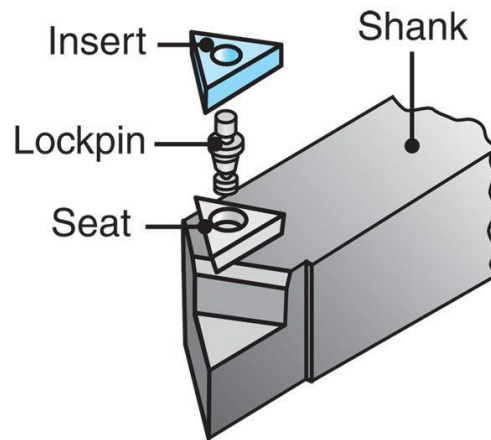
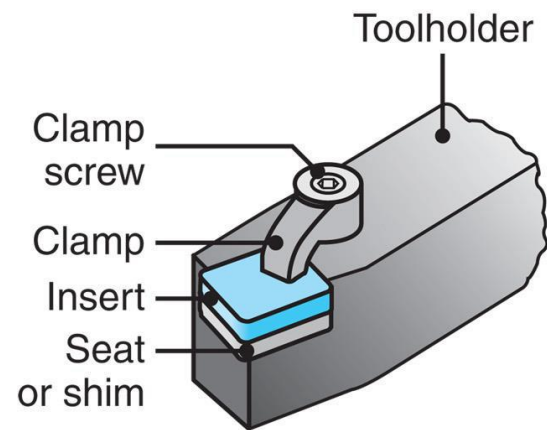
(l) Knurling

Utensili

Materiali degli utensili

Tool materials	General characteristics	Modes of tool wear or failure	Limitations
High-speed steels Acciai	High toughness, resistance to fracture, wide range of roughing and finishing cuts, good for interrupted cuts	Flank wear, crater wear	Low hot hardness, limited hardenability, and limited wear resistance
Uncoated carbides	High hardness over a wide range of temperatures, toughness, wear resistance, versatile, wide range of applications	Flank wear, crater wear	Cannot use at low speeds because of cold welding of chips and microchipping
Coated carbides	Improved wear resistance over uncoated carbides, better frictional and thermal properties	Flank wear, crater wear	Cannot use at low speeds because of cold welding of chips and microchipping
Ceramics	High hardness at elevated temperatures, high abrasive wear resistance	Depth-of-cut line notching, microchipping, gross fracture	Low strength and low thermomechanical fatigue strength
Polycrystalline cubic boron nitride (cBN)	High hot hardness, toughness, cutting-edge strength	Depth-of-cut line notching, chipping, oxidation, graphitization	Low strength, and lower chemical stability than ceramics at higher temperature
Diamond	High hardness and toughness, abrasive wear resistance	Chipping, oxidation, graphitization	Low strength, and low chemical stability at higher temperatures

Utensili

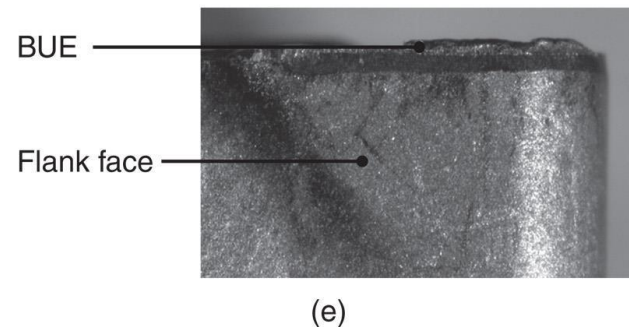
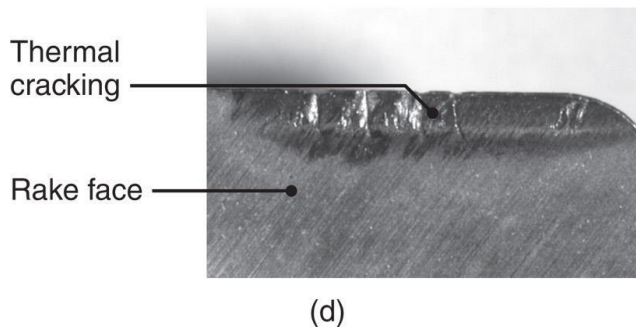
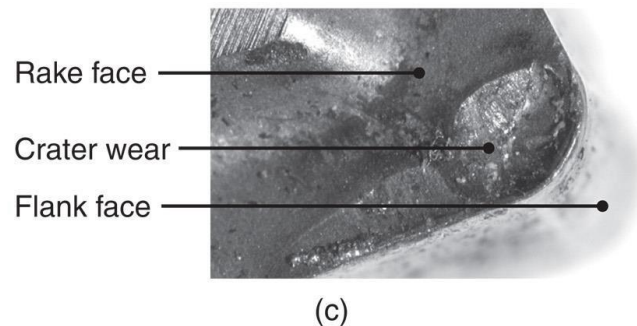
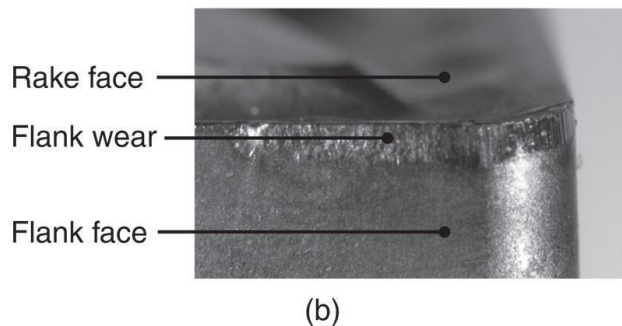
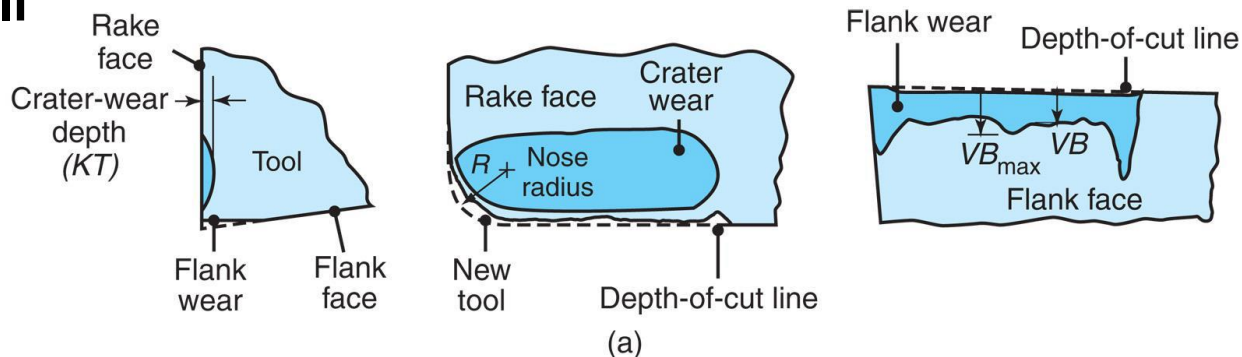


[Video: comparazione inserti](#)

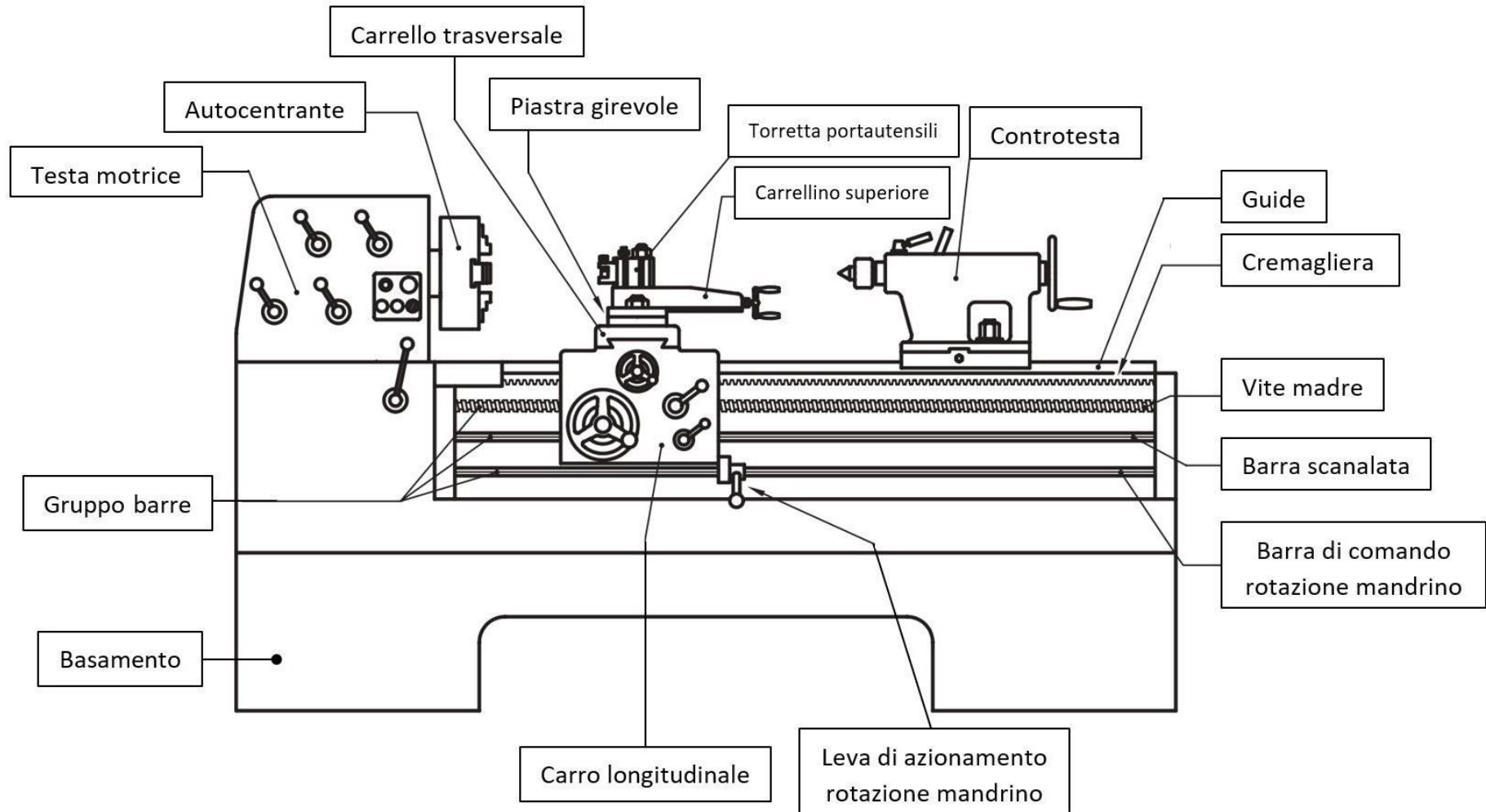
Utensili

Usura degli utensili

Le considerazioni sugli utensili valgono anche per fresatura e foratura

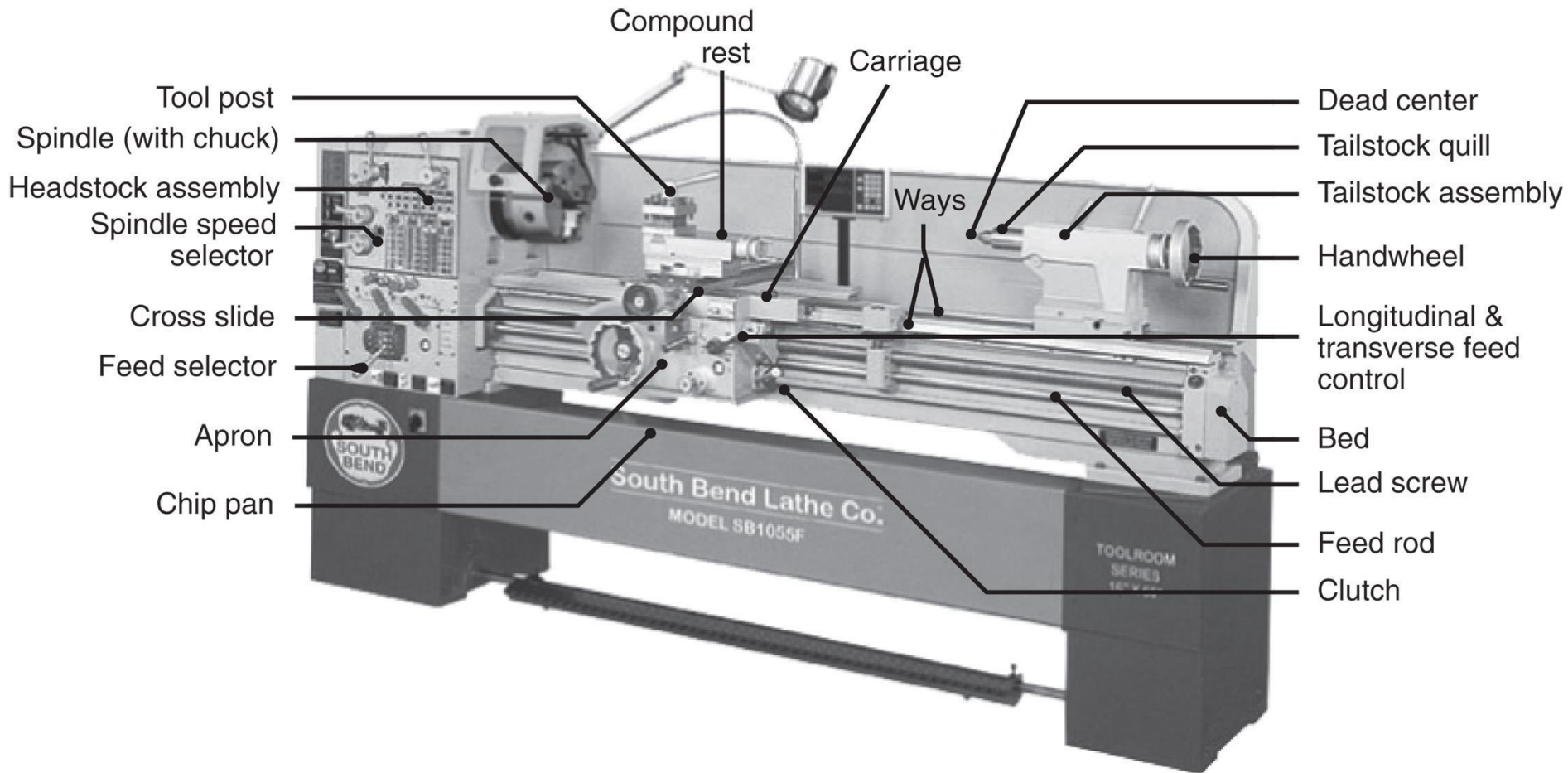


Tornitura



<https://www.techmec.it/tornio-parallelo-manuale/>

Tornitura

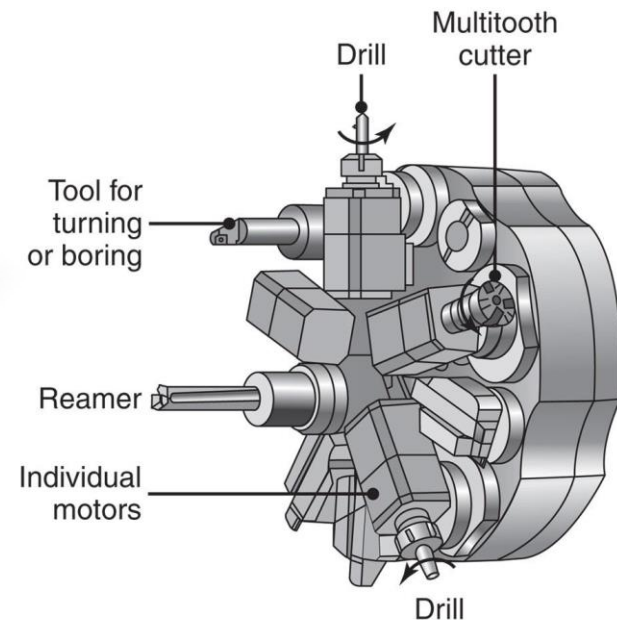


Tornitura



[Video: tornio CNC](#)

Attenzione: nel video vengono mostrate anche operazioni di fresatura.



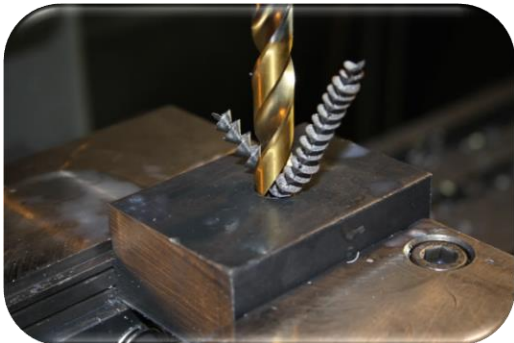
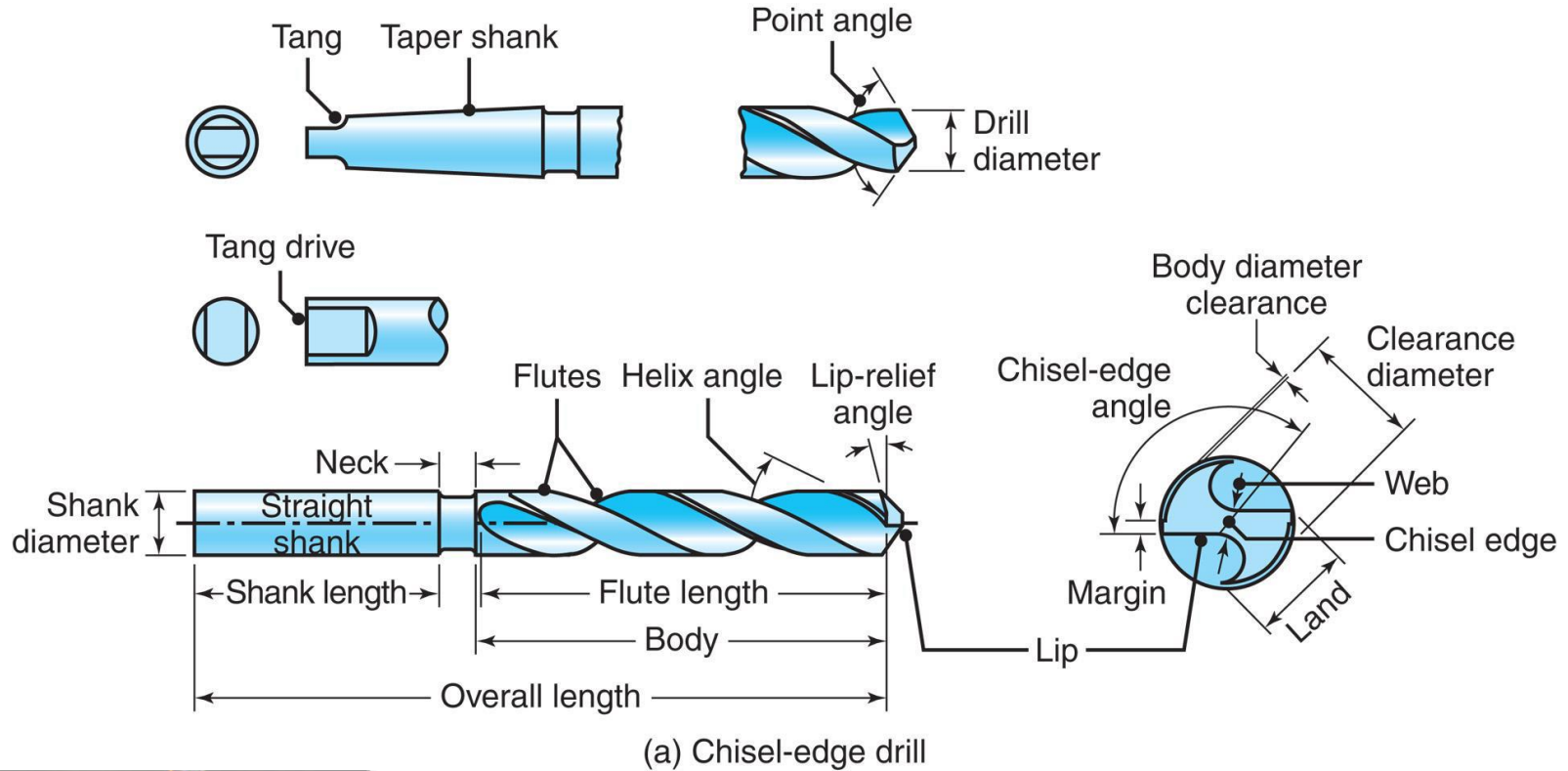
Tornitura



Fori trasversali
ottenuti mediante
foratura successiva

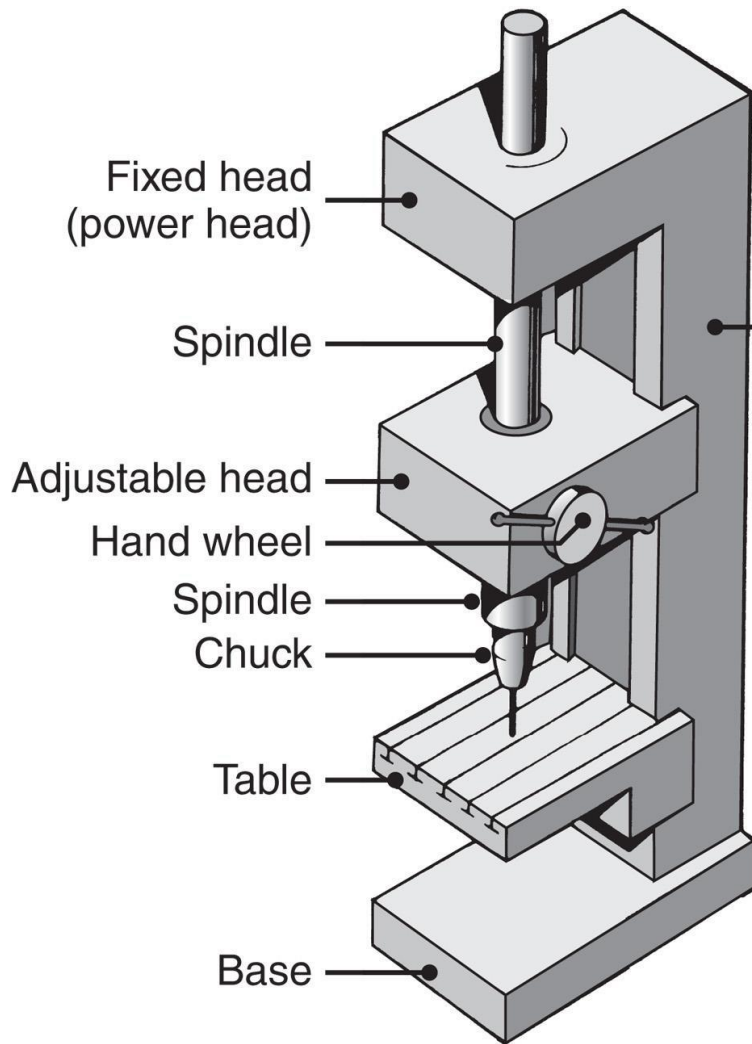


Foratura

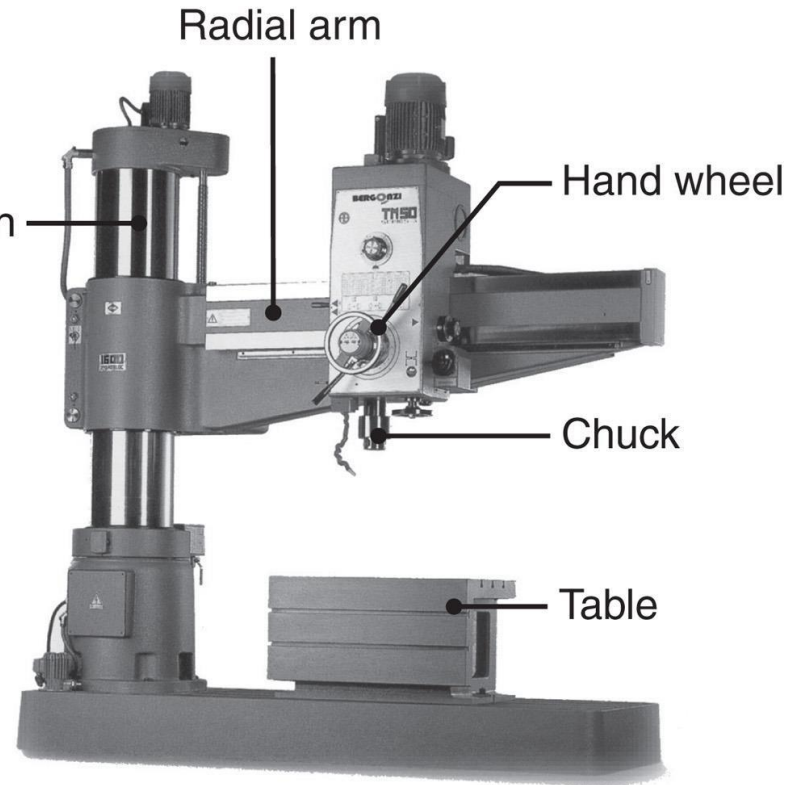


[Video: foratura](#)

Foratura



(a)



(b)

Fresatura



[Video](#)

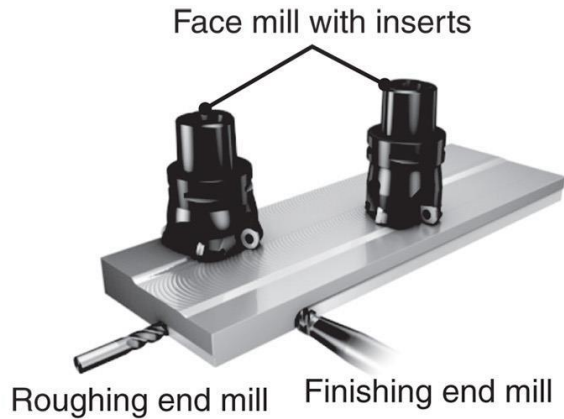
[Video: utensili](#)

Fresatura

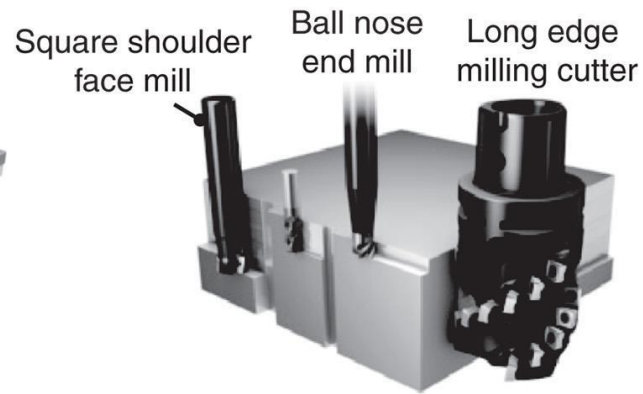


Approfondimento: [Sandvik Coromant](#)

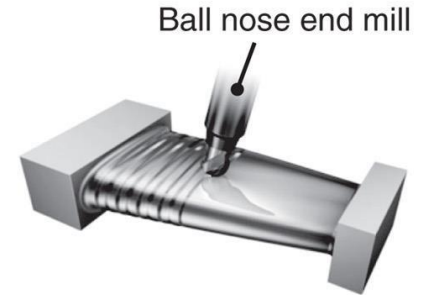
Fresatura



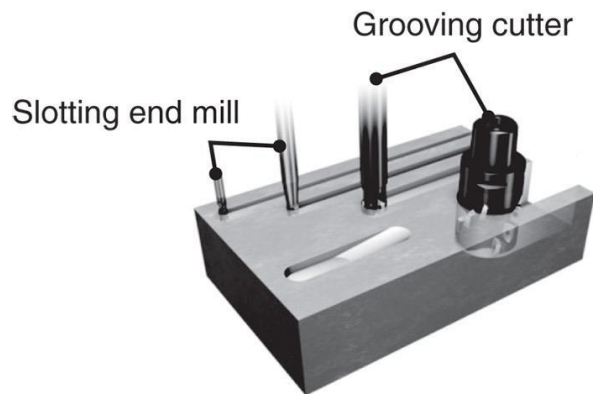
(a) Face milling



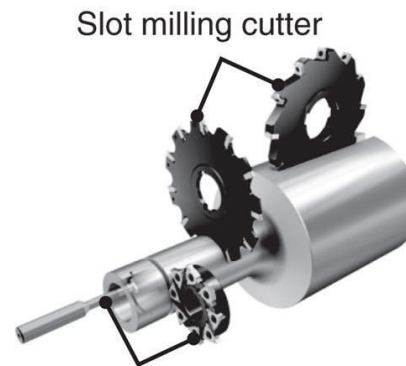
(b) End or shoulder milling



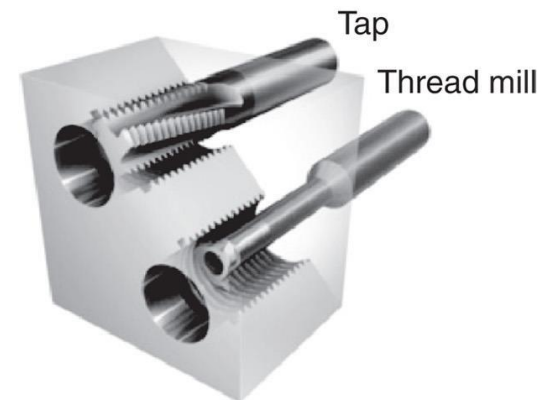
(c) Profile milling



(d) Slot milling



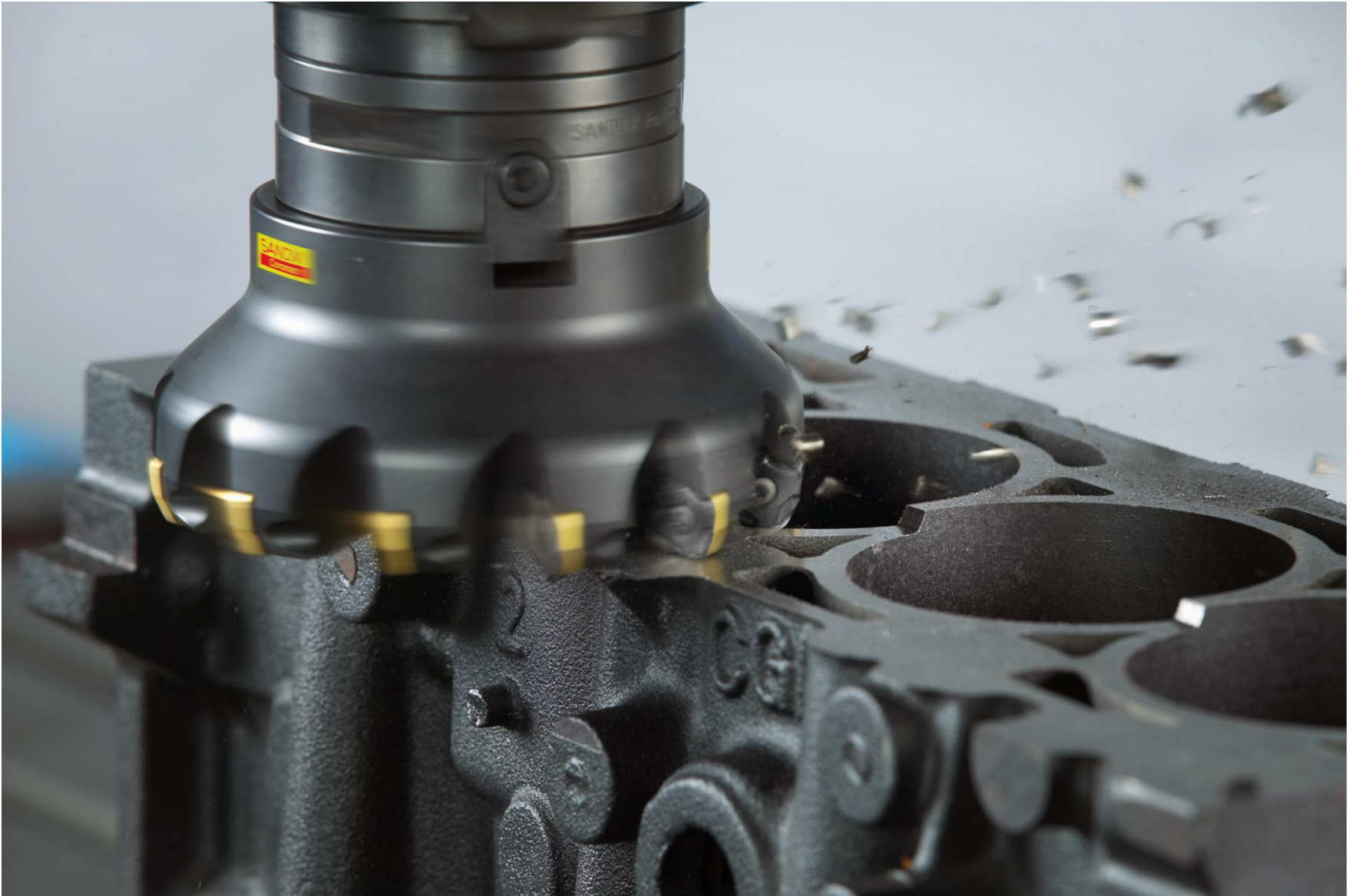
(e) Slot and groove milling



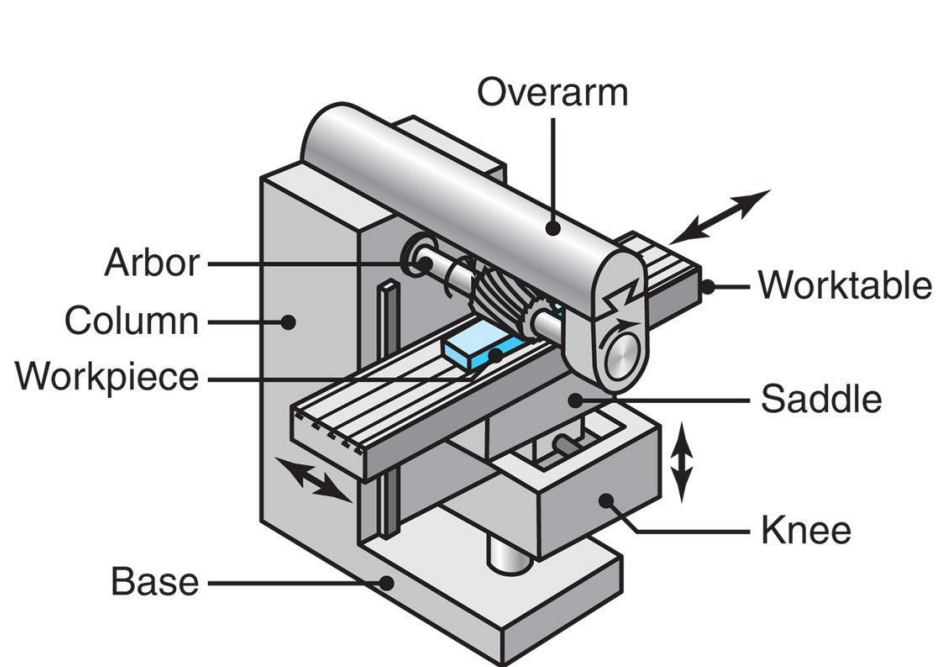
(f) Thread milling and tapping

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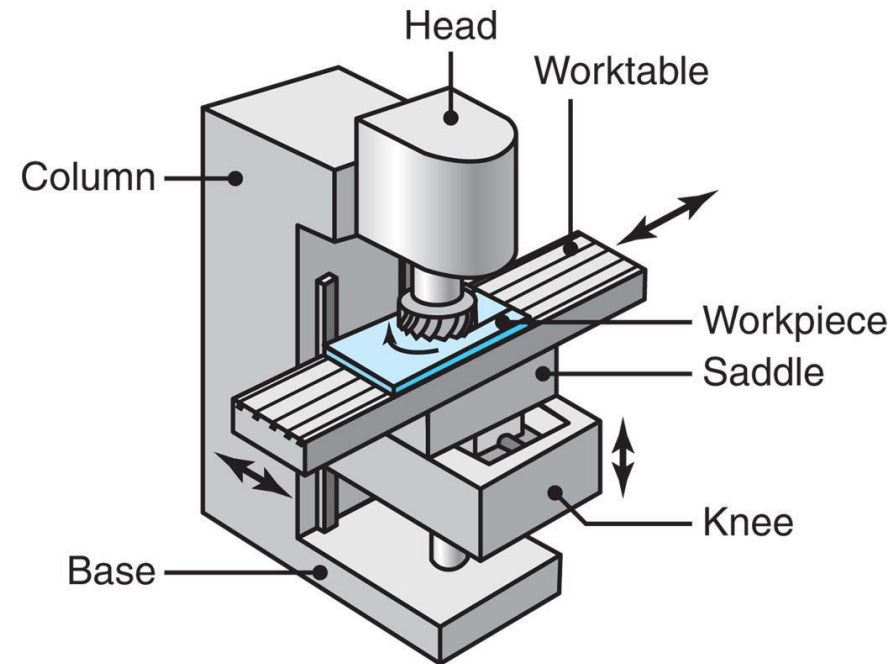
Fresatura



Fresatura



Fresatrice a mandrino
orizzontale

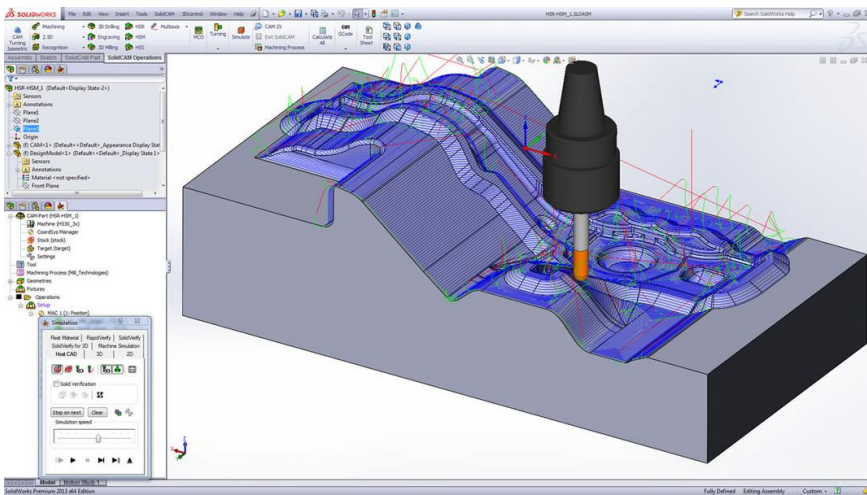


Fresatrice a mandrino
verticale

Fresatura



Fresatura

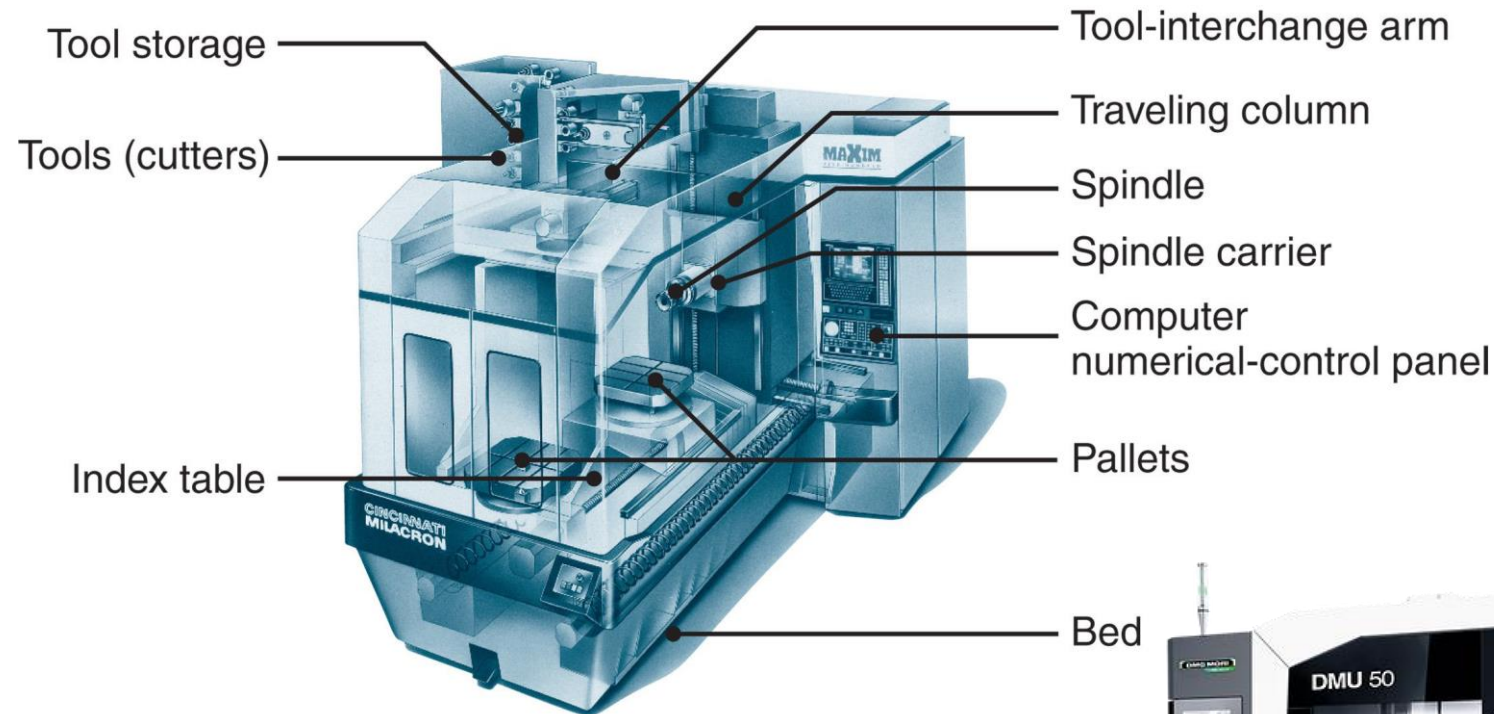


(Start cutting path id: path29632)
(Change tool to Cylindrical cutter)

```
G00 Z5.000000  
G00 X33.655106 Y11.817060  
  
G01 Z-1.000000 F100.0(Penetrare)  
G01 X247.951560 Y11.817060 Z-1.000000 F400.000000  
G01 X247.951560 Y30.935930 Z-1.000000  
G01 X106.963450 Y30.935930 Z-1.000000  
G03 X106.587404 Y32.243414 Z-1.000000 I-7.576860 J-1.471361  
G03 X105.974610 Y33.458880 Z-1.000000 I-6.445333 J-2.487300  
G03 X104.697090 Y35.083261 Z-1.000000 I-7.601246 J-4.663564  
G03 X103.141830 Y36.435630 Z-1.000000 I-10.087550 J-10.030472  
G03 X102.969400 Y38.107779 Z-1.000000 I-20.252028 J-1.243405  
G03 X102.369430 Y39.685740 Z-1.000000 I-3.842423 J-0.557919  
G03 X100.419761 Y41.664361 Z-1.000000 I-6.181245 J-4.140917  
G02 X98.333794 Y43.482560 Z-1.000000 I7.045018 J10.188229  
G02 X95.783544 Y47.017541 Z-1.000000 I9.647185 J9.647199  
G02 X94.101654 Y51.024620 Z-1.000000 I28.957871 J14.510988  
G03 X92.872672 Y54.561719 Z-1.000000 I-340.631289 J-116.371936  
G03 X91.674043 Y58.106100 Z-1.000000 I120.624220 J46.152281
```



Fresatura



[Video: Fresatura
5 assi](#)



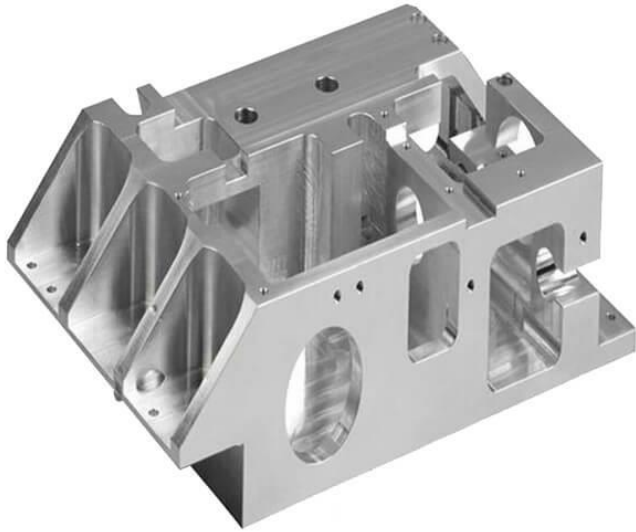
Esercizio: prodotti

Cliccare su [questo link](#) per raggiungere la slide condivisa.

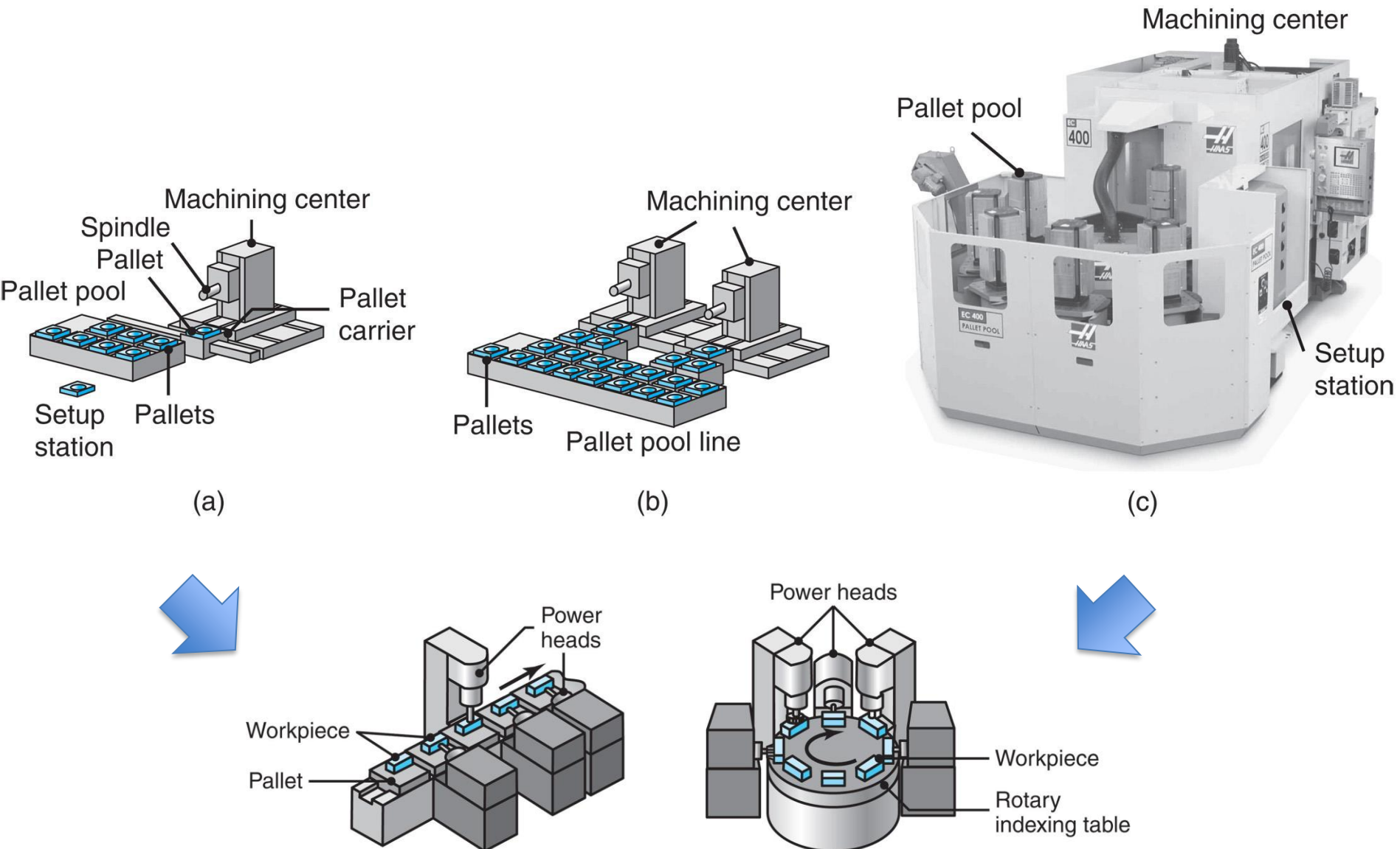
Inserire un'immagine rappresentativa di un prodotto ottenuto mediante asportazione di truciolo. Lo stesso link è accessibile tramite il QR code:



Fresatura



Fresatura



Industria 4.0





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Dipartimento di Ingegneria
"Enzo Ferrari"

Domande?

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